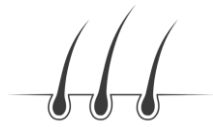


Hair Loss Prediction Using Deep Learning

A hybrid system that classifies Norwood stage from scalp images and predicts hair-loss risk from survey data, then fuses both to deliver personalized recommendations.

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Background About The Project



Problem

Clinics rely on manual photo reviews and patient forms; staff evaluate images, contact patients, and provide feedback.

Process is slow and resource-heavy, delaying treatment recommendations.

Current workflow under-utilizes technology to align care with a patient's hair-loss stage.

Goal

Automate initial intake steps using ML: analyze patient photos and estimate hair-loss stage (e.g., Norwood).

Accelerate evaluations, reduce manual effort, and tailor feedback to the patient's current condition.

Success Criteria

≥ 70% accuracy for hair-loss level prediction.

30% reduction in time for initial evaluations via a digitized intake workflow.

≥ 5% increase in customer conversion due to efficiency and satisfaction.

Norwood Scale

1



2



3



3

vertex



4



5



6



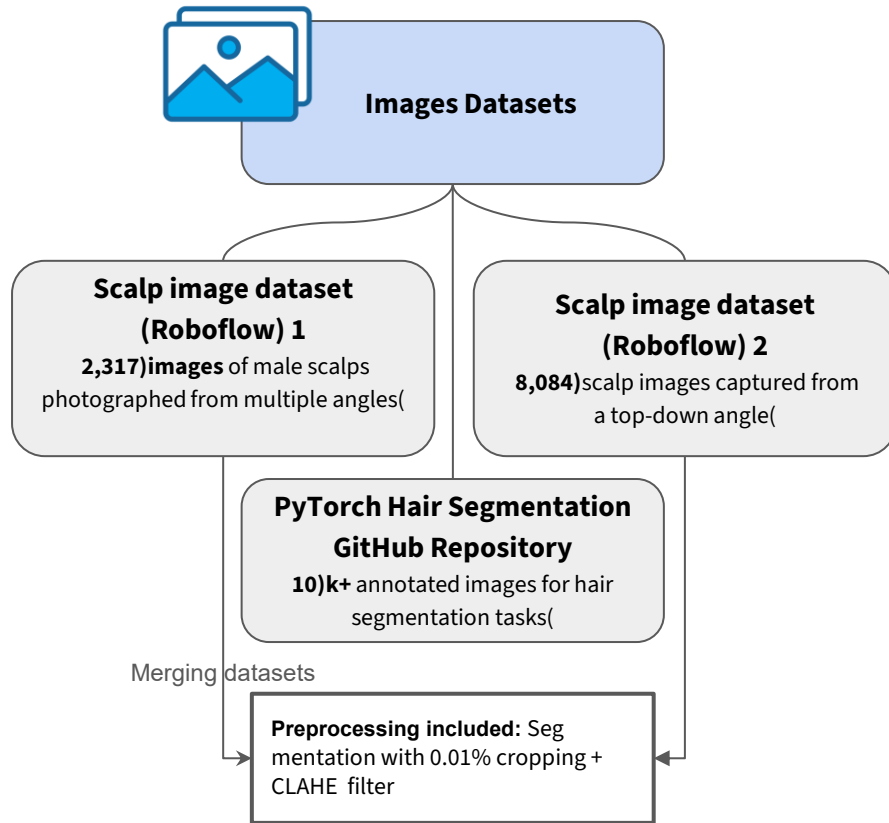
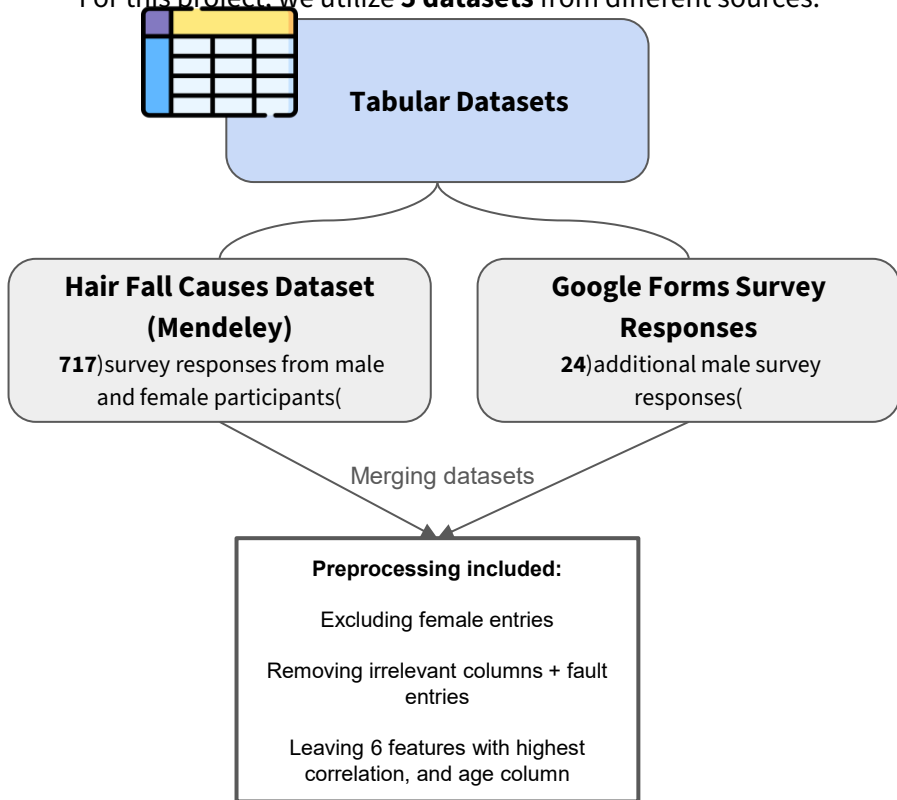
7



Data Sources & Preparation



For this project, we utilize **5 datasets** from different sources:



1



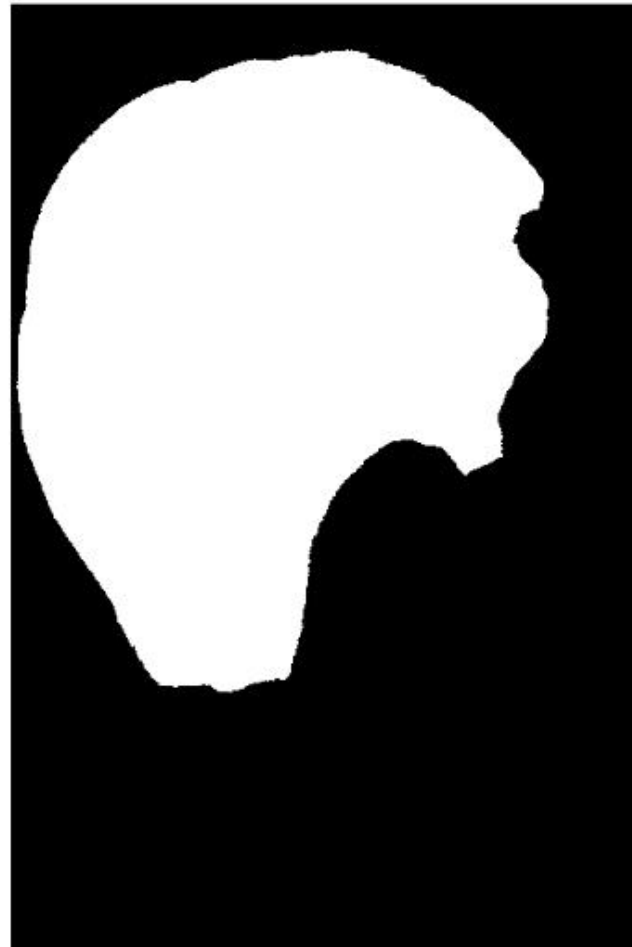
PyTorch Hair Segmentation

GitHub Repository

10)k+ annotated images for hair
segmentation tasks(

**Deeplab +
ResNet50**

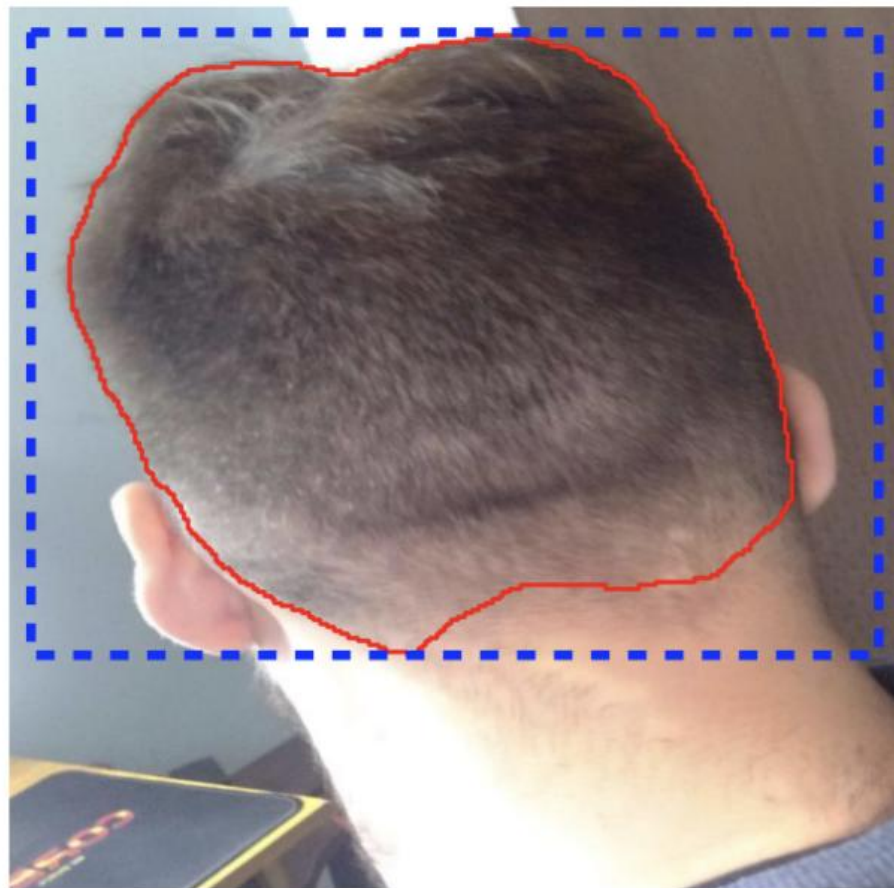
2



1



2



Modeling Approach



Tabular Data:

- Tested **8 Machine Learning models** dna tseroF modnaR ,NNK ,(VChcraeSdirG)NNK sa hcus .tsooBGX
- The goal is to predict **whether a person has hair fall problem**.yevrus eht no desab

Images Data:

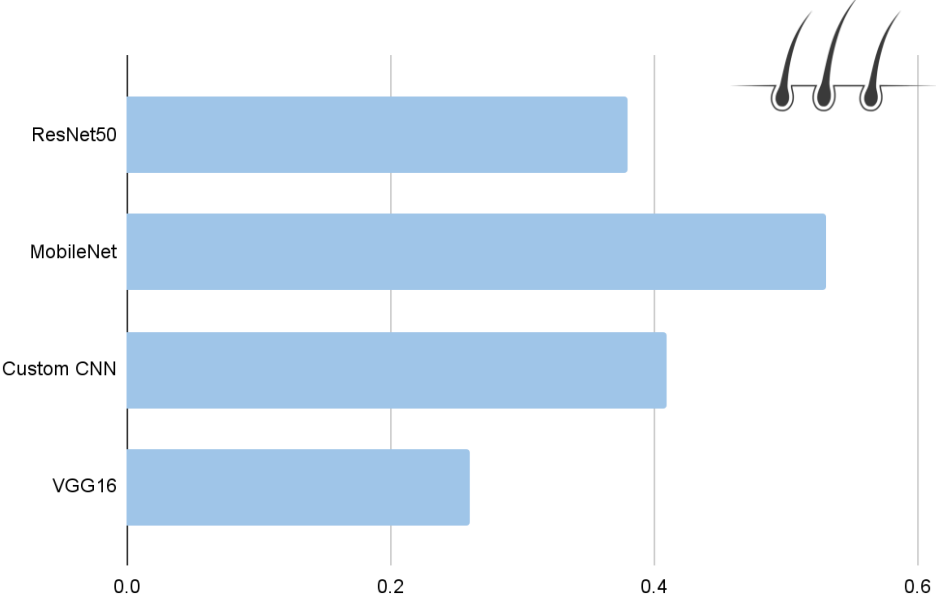
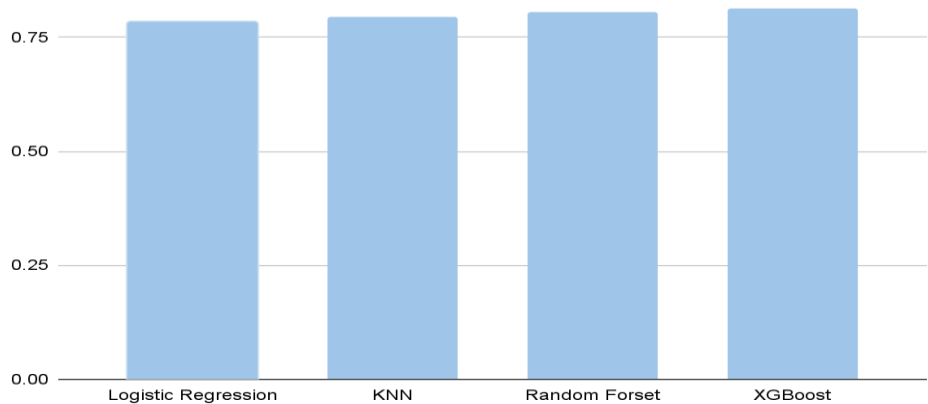
- Trained **13 CNN based models** dna egraL-3VteNeliboM ,0B-teNtneiciffE ,18teNseR gnidulcni , otni segami yfissalc ot yniT-tXeNvnoC**Norwood scale (Levels (1-7**

The goal is to use both classification in order to tailor personalized **tnemtaert dna evitatneverp .ylgnidrocca serusaem**

Initial Results & Evaluation Before Improvements

Images Data:

Initially, **MobileNet** demrofrep tuo tset a gniveihca ,sledom rehto eht llef siht ,revewoH .68% fo ycarucca derolpxe ew os ,tegrat ruo fo trohs evorpmi ot sdohtem evitanretla .ycarucca



Tabular Data:

The model that provided the best results is **XGBoost**81% fo ycarucca htiw



We decided to try more models, **merge our existing dataset with an additional one** eht htiw ,noitacifissalc emas**apply segmentation** dna ,noiger riah eht etalosi ot **apply CLAHE** ot llew sa .yltnacifingis ycarucca eht devorpmi ew ,ycarucca noitciderp eht evorpmi

Model Name	Accuracy	F1 Score
ResNet18	0.86	0.7668
MobileNet-VeagraL-3	0.86	0.75
EfficientNet-B0	0.85	0.75
ConvNext-Tiny	0.85	0.75

BEFORE



AFTER



Deployment - Tkinter based UI



Hugging Face

deeplabv3-hair-segmentation
efficientnet-b0_norwood_classifier
xgboost_hair_fall_classifier
mobilenet_v3_large_cropped_clahe_norwood_classifier
convnext_tiny_clahe_norwood_classifier
Resnet18
densenet121



USER INPUT



Yes/No Form Image

loading_screen.py

intake_screen.py

processing_screen.py

results_screen.py

main.py

OUTPUT 1 & 2



Yes hair fall /
No hair fall

AND

1 → 2 → 3 → 4 → 5 → 6 → 7



Norwood
Scale Levels
1-7

OUTPUT 3



**Personalized
recommendations**

THANK YOU FOR LISTENING

Your hairline is more than just hair - it's confidence, presence, and the best version of you. Don't settle for less.

